Classification of States

Definition (Accessibility).
State j is **accessible** from state i, denoted by $i \to j$, if

$$p_{ij}^{(n)} > 0 \quad \text{for some } n \geq 0.$$

Note. $p_{ij}^{(0)} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$

Remark.

- Accessibility is not a one-step concept: p_{ij} may be zero while $p_{ij}^{(n)} > 0$ for some $n > 1$.
- It is not symmetric: $i \to j$ does not imply $j \to i$.
- Since $p_{ii}^{(0)} = 1$, every state is accessible from itself.

Definition (Communication).
States i and j **communicate**, denoted by $i \longleftrightarrow j$, if

$$i \to j \qquad \text{and} \qquad j \to i.$$

Note. Communication is an **equivalence** relation.

Definition (Communication Class).
A **communication class** is a **maximal** set $\mathscr{C} \subseteq S$ whose states communicate with one another.

Note. Communication classes form a partition of the state space S.